2026 / 03 / 06

How Did **Donald Trump's Tweets** Affect **Industry-Level Stock Returns?**

Summary

Return

- Strong short-term abnormal returns in **Financials and Semiconductors**
- **News & Publishing** shows the only significant long-term decline
- Most price effects revert within days

Volatility

- Short-term spikes in **High-Tech and information-sensitive sector**. Effects concentrated rather than market-wide

- No systematic long-term persistence

Literature Review

Firm-level Effects

- Company-specific tweets generate significant **abnormal returns (AR)** on the same day.
- Negative tweets tend to produce **delayed negative effects**.
- Tweets act as **information shocks**, with markets reacting faster than traditional media.

Market-level Effects

- Tweets increase overall market **volatility** and **trading volume**.
- Trade- and politically-related tweets are more likely to trigger **negative market reactions**.
- The effects are mainly concentrated within **short event windows**.

Methodology: EVENT STUDY

Resources *:*

Copenhagen Business School, May 15, 2019, "President Trump, his Tweets and the Stock Market——A study of how Trump's tweets affect targeted companies' stock prices" J
ournal of International Money and Finance, June, 2025 "Signal in the noise: Trump tweets and the currency market"

Decision Support Systems (2021)"Do President Trump's tweets affect *financial markets?"*

Hypotheses

GENERAL HYPOTHESIS

Trump's tweets mentioning specific companies generate **significant abnormal influence on stock returns** in related industries.

SUB-HYPOTHESIS 1

Long-term effects are weaker than short-term effects. Over time, the impact largely reverts to normal levels.

SUB-HYPOTHESIS 2

The impact is mainly concentrated in the short term, and significantly differs across industries.

Contents

1.Data Description

Data Source

Stocks Data

- **EODHD** : https://eodhd.com/
- Sample Period: **2009–2021**
- All dates and timestamps are reported in **UTC**.
- **50** companies in **10** industries.

Tweets Data

- **Trump Twitter Archive V2** : https://www.thetrumparchive.com/?resultssortOption=%22Latest%22
- Sample Period: **2009–2021**
- All dates and timestamps are reported in **UTC**.

Stock Data Preprocessing

Data Preparation

Minutely Data

Restricted to regular trading hours (ET 9:30–16:00, UTC converted)

Missing minutes filled forward with last observed price

Volume set to 0 for filled minutes

Returns computed on the cleaned minute series

Hourly Data

Aggregated from minute data

Arithmetic mean of prices within each trading hour

Volume summed within each hour

Hourly returns computed from mean close

Daily Data

Open: first trade of the day

Close: last trade of the day

Volume: total daily sum

Daily returns computed from close

Event Window Selection Algorithm

Input: Tweet timestamp (UTC)

Step 1: Align event to nearest valid trading period

If outside trading hours → snap to closest session anchor

Step 2: Select fixed window

N trading periods before

M trading periods after

Step 3: Extract stock data at those trading timestamps

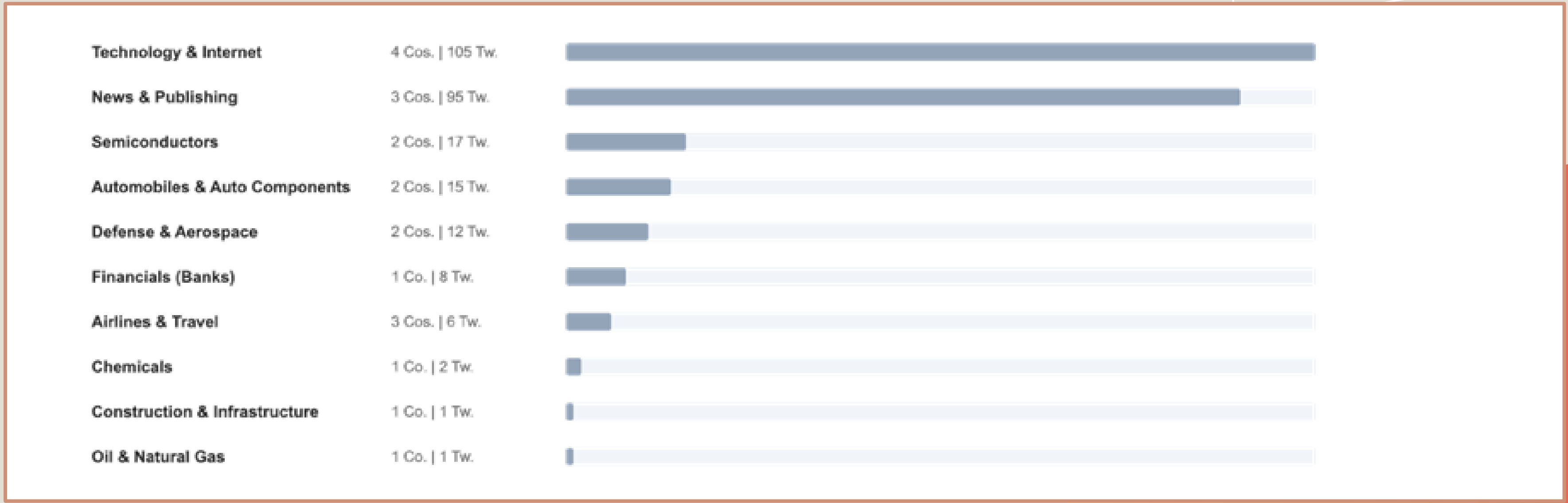
Step 4: (Optional) Aggregate to industry level

Result: Clean, trading-time-aligned event panel excluding weekends and holidays.

Tweets Data Preprocessing

Raw Dataset → Remove Retweets → Filter Time → Dictionary&Match → Hourly Filter

Raw Dataset	Remove Retweets	Filter Time	Dictionary&Match	Hourly Filter
56,571	46,694	14,369	262	228



Industry & Firm Selection

INDUSTRY CRITERIA

Policy Expectation Revision Sensitivity

We select industries whose valuations are highly sensitive to policy expectation revisions. Presidential tweets affect firm value by shifting expectations regarding:

- Trade policy
- Regulation
- Fiscal spending
- Geopolitical risk
- Monetary conditions

Tweets → Expectation Revision → Stock Returns

TRADE & GLOBAL POLICY EXPOSURE

Semiconductors | Automobiles | Airlines & Travel | Chemicals

REGULATORY & POLICY DEPENDENCE

Technology & Internet | Financials (Banks) | Media & Publishing

FISCAL & GOVERNMENT SPENDING

Construction & Infrastructure | Defense & Aerospace

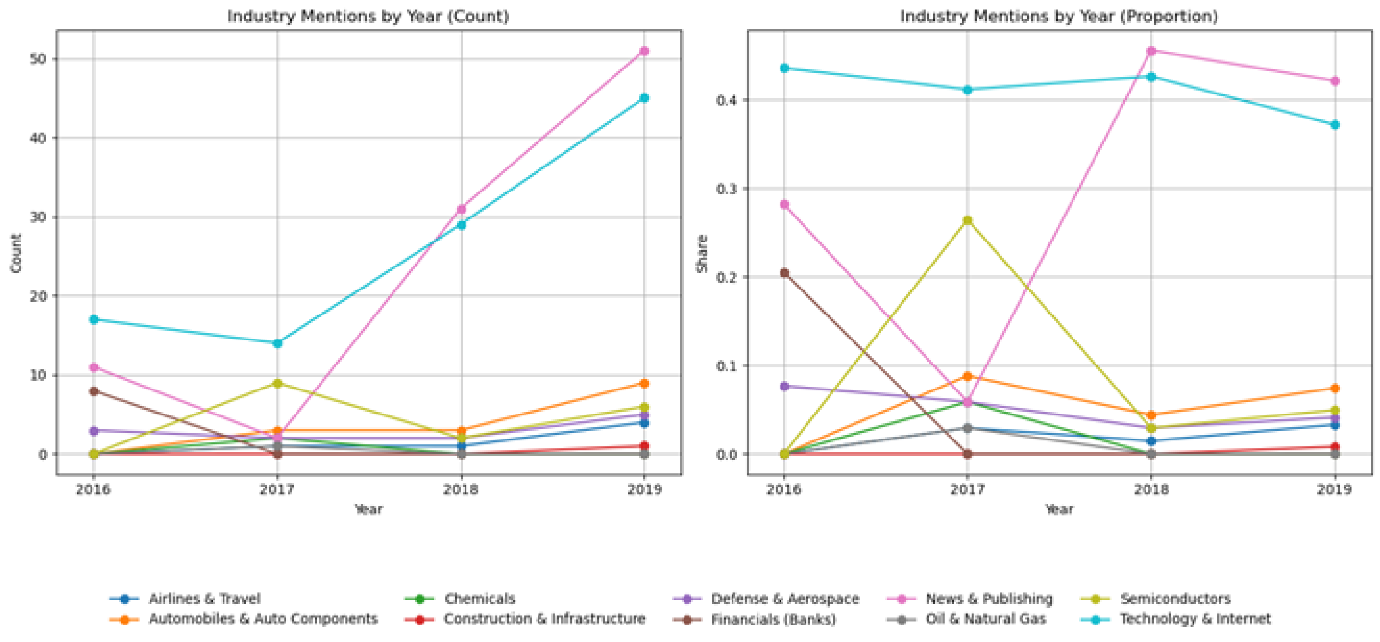
COMMODITY & GEOPOLITICAL EXPOSURE

Oil & Gas

FIRM CRITERIA

- **Large-cap S&P 500 firms**
- **High liquidity and data availability**

Tweets Time Distribution



Structural shift in 2018
Clear redistribution of industry weight within the selected sample.

Concentration in Tech & News
By 2019, ~80% of mentions come from these two sectors.

Composition-driven share changes
Share movements reflect sample structure, not the full market.

2.EVENT STUDY Methodology

Event Definition and Event Windows

- **Event study** measures the impact of **a specific event** on asset by comparing actual performance with expected performance **around the event window.**

228 Events

Tweet Eligibility

Mentions directly include:

-Company Name

-Alias

-Abbreviation

Multi-Company Tweets

Same Industry:

One single event

Across Industries:

Split by industry

Shared Event ID

Duplicate Filtering

If multiple tweets target the same industry within:

Less than 1 hour:

Only keep first tweet

Event Windows

Captured market reactions:

Short-Term Window

[-60min,+60min]

Long-Term Window

[-5day,+5day]

EVENT STUDY Calculations

RETURN

Definition: Percentage change in stock price.

Formula:

$$R_t = \frac{P_t^{close} - P_{t-1}^{close}}{P_{t-1}^{close}}$$

$$AR_t = R_t - E[R_t]$$

Key Metrics: AAR_t, $CAAR[t_1,t_2]$

VOLATILITY

Definition: Day-to-day price fluctuation, as a proxy for realized variance.

Formula:

$$RV_t = r_t^2$$

Key Metrics:

$$\Delta RV = RV_{post} - RV_{pre}$$

T-test :

$$H_0 : \text{Abnormal Effect} = 0$$

3. Analysis & Findings

Analysis 01

Return

Volatility

Volume

Year/Quarter/Month

Short-term vs. Long-term Return Event Study

	short-term	long-term
Data Frequency	Minutely	Daily
Event Window	[0,+60 mins]	[-5,+5 days]
Estimation Window	[-60,-1 mins]	[-250, -10 days]
Benchmark Model	Mean-adjusted	CAPM (OLS)

Return Definition

$$R_{i,j,t} = \frac{P^{close}_{i,j,t} - P^{close}_{i,j,t-1}}{P^{close}_{i,j,t-1}}$$

Short-term (Minutely, $t \in [-60,\ +60]$ min)

$$\hat{\mu}_{i,j} = \frac{1}{60} \sum_{t=-60}^{-1} R_{i,j,t}, \quad \text{(pre-tweet mean, estimation window)}$$

$$AR_{i,j,t} = R_{i,j,t} - \hat{\mu}_{i,j}, \quad t \in [-60,\ +60]$$

Long-term (Daily, $t \in [-5,\ +5]$)

$$R_{i,j,t} = \alpha_{i,j} + \beta_{i,j} \cdot R_{m,t} + \varepsilon_{i,j,t}, \quad t \in [-250,\ -10]$$

$$\hat{R}_{i,j,t} = \hat{\alpha}_{i,j} + \hat{\beta}_{i,j} \cdot R_{m,t}, \quad t \in [-5,\ +5]$$

$$AR_{i,j,t} = R_{i,j,t} - \hat{R}_{i,j,t}$$

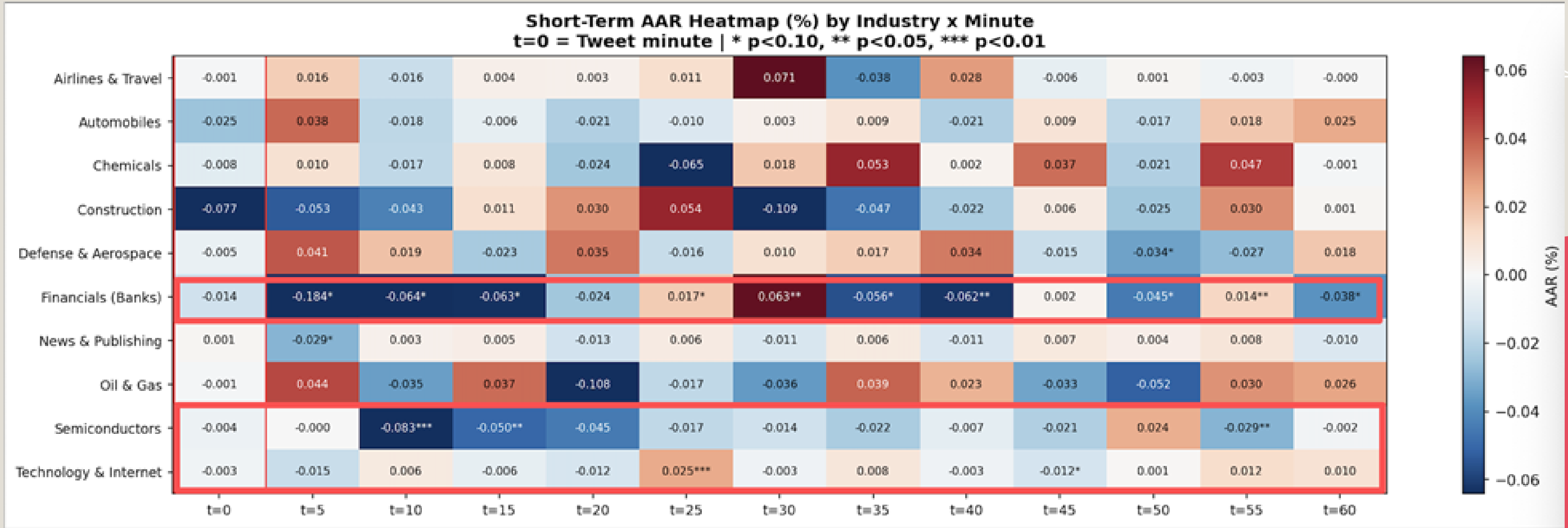
Aggregation (both models)

$$AAR_{j,t} = \frac{1}{N_j} \sum_{i=1}^{N_j} AR_{i,j,t} \quad (N_j = \text{number of tweet events in industry } j)$$

$$CAAR_j[t_1, t_2] = \sum_{t=t_1}^{t_2} AAR_{j,t}$$

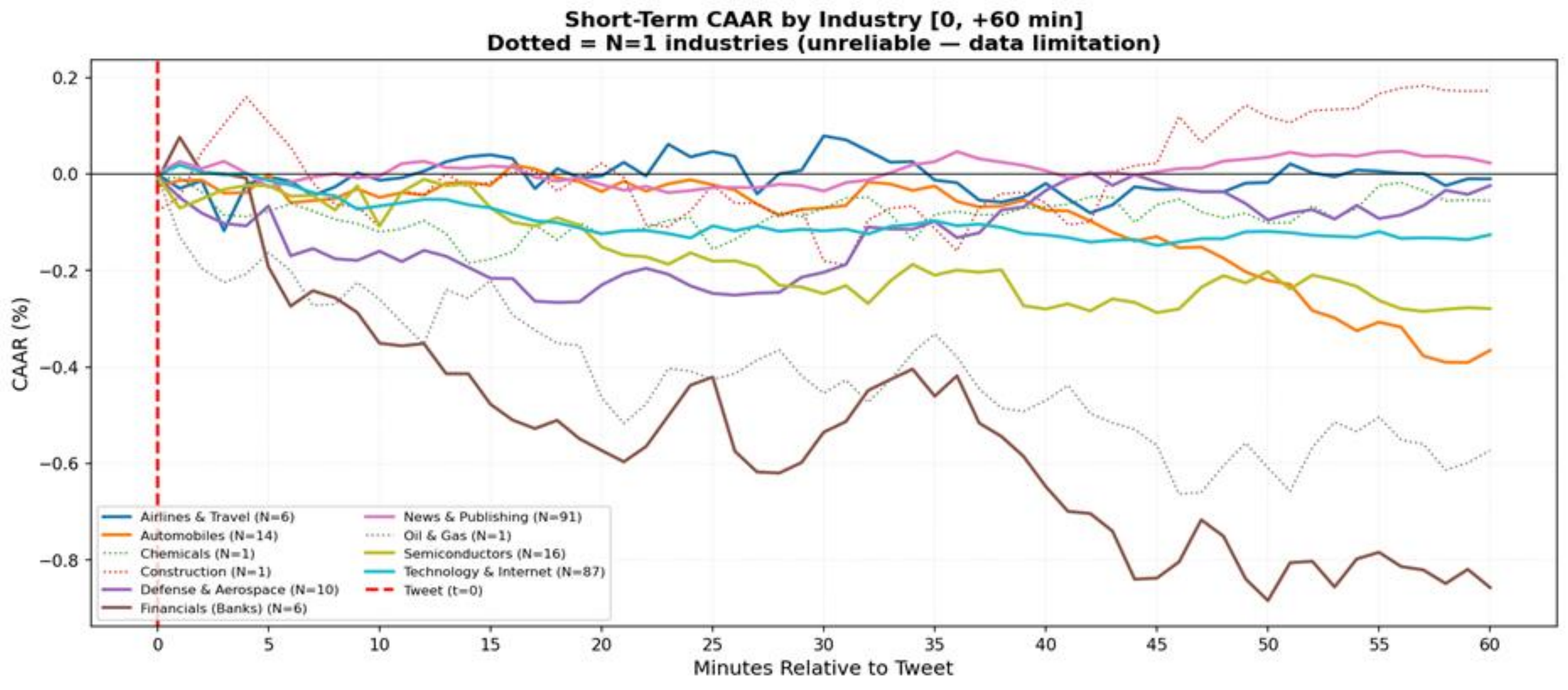
Short-term Reaction: Financials and Semiconductors React Within 15 Minutes

Financials (Banks) and **Semiconductors** show the strongest and most statistically significant immediate reactions (within 15 minutes of the tweet), **Technology** reacts significantly around 25 minutes, while most other industries show no significant short-term AAR.



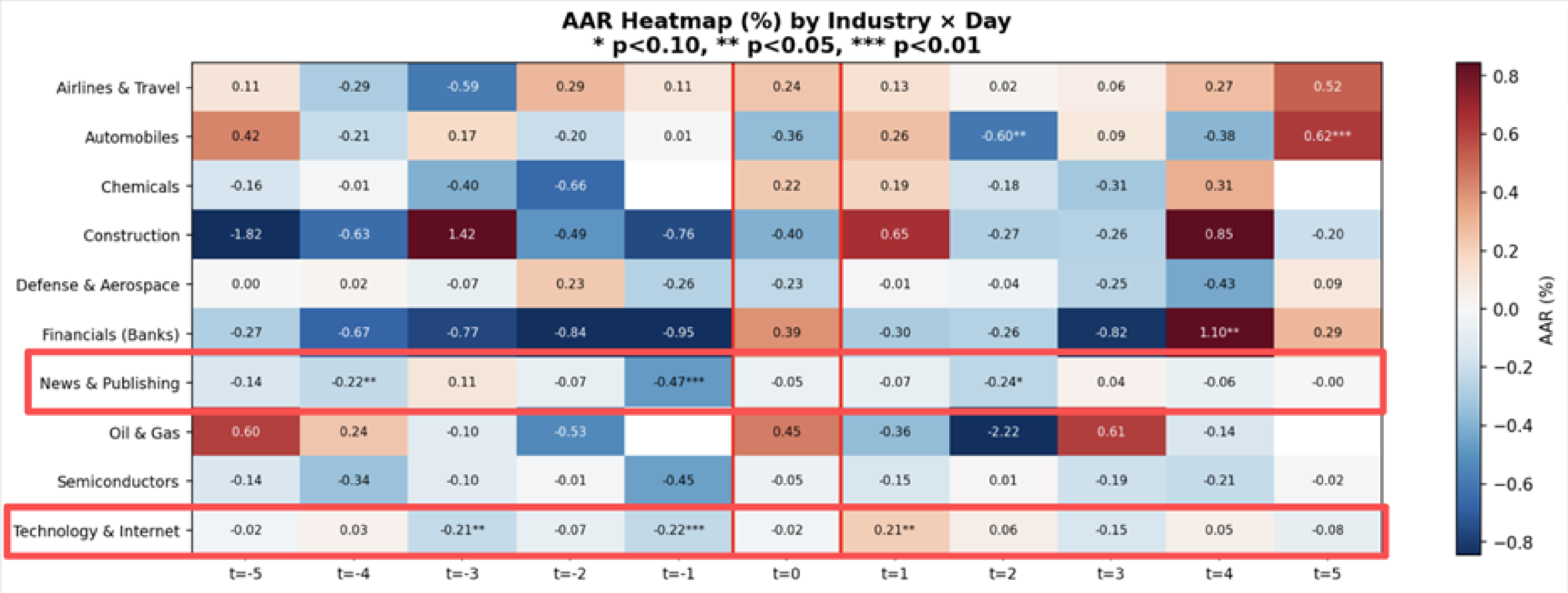
Short-term Cumulative Impact: Financials Fall Immediately, Automobiles Lag

In terms of magnitude, **Financials** is clearly the most significantly affected industry, followed by **Automobiles** and **Semiconductors**, while other sectors show comparatively smaller or temporary movements.



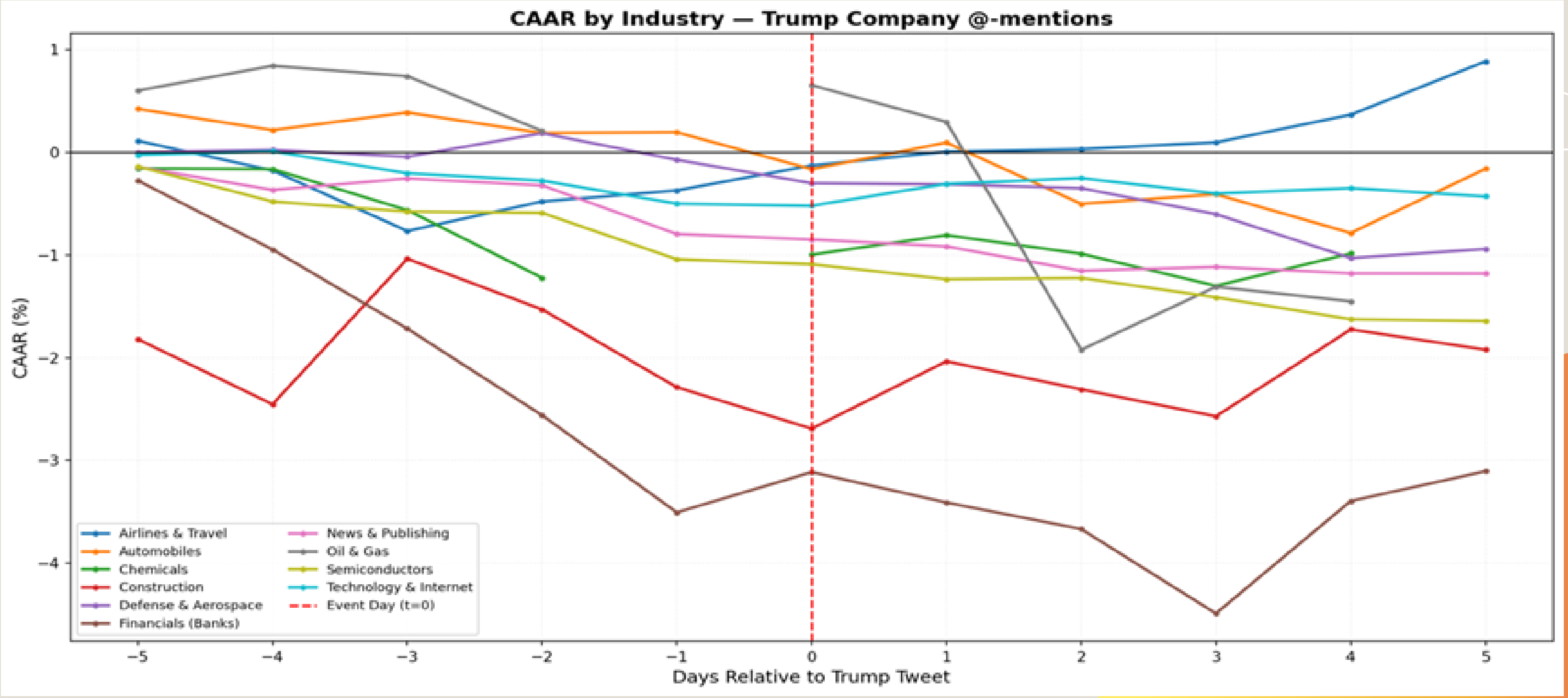
Long-term Reaction: Pre-tweet Signals and Post-tweet Persistence

Significant daily average abnormal returns are concentrated in **News & Publishing** and **Technology & Internet**, particularly around t = -1 and the immediate post-event day.Most other industries exhibit only isolated single-day movements without consistent multi-day patterns.



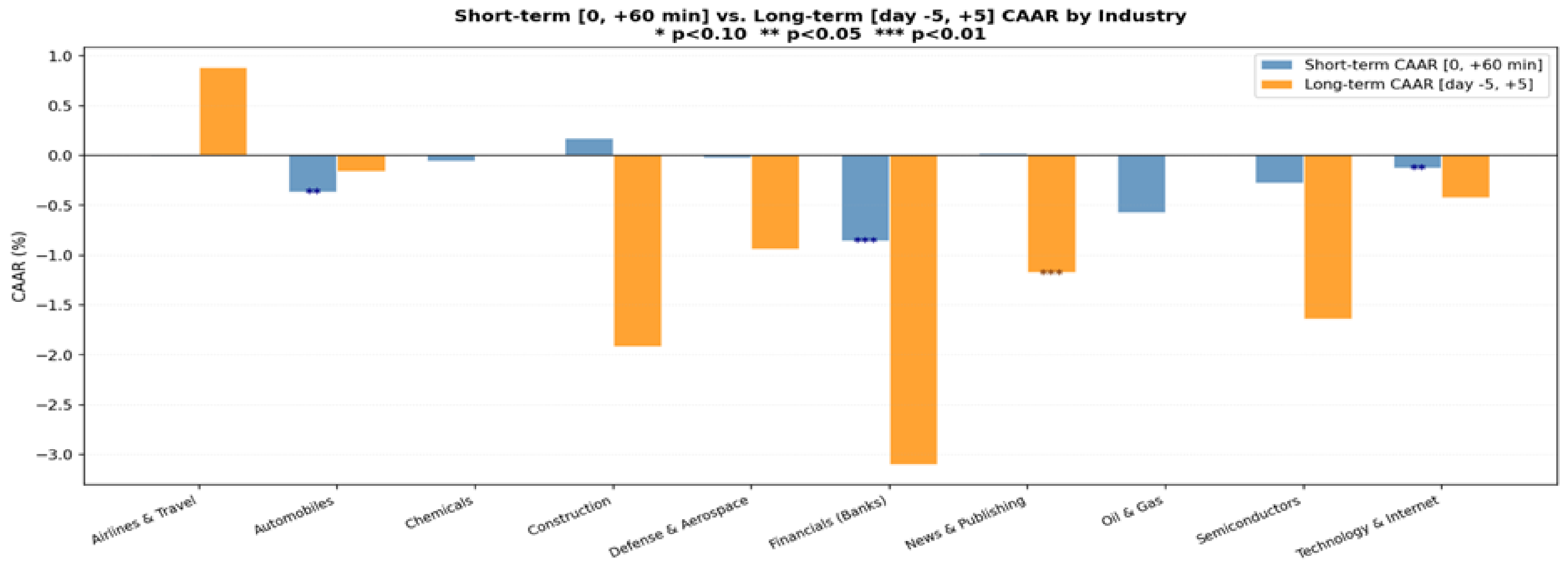
Long-term Cumulative Impact: News & Publishing Suffers the Most Significant Loss

Trump's company @-mentions generate the most visible cumulative declines in **Financials and Construction**, while **News & Publishing** shows the only statistically significant long-term loss. **Airlines** stands out as the only industry with a clear positive cumulative response, whereas most other sectors exhibit moderate downward movements.



Two Transmission Mechanisms: Immediate Shock vs. Slow Diffusion

Trump's company @-mentions trigger heterogeneous industry responses, with **Financials** and **Automobiles** exhibiting immediate and persistent reactions, while **News & Publishing** shows delayed but statistically significant long-term losses.These patterns indicate the presence of two distinct transmission mechanisms: immediate shock versus gradual diffusion.



Findings 01

Return

Short-term

- In the short term, industry reactions are concentrated in **Financials and Semiconductors**, which exhibit immediate and statistically significant abnormal returns within minutes of the tweet.
  Most other sectors show limited or temporary short-term responses, indicating that immediate market reactions are selective rather than broad-based.

Long-term

- In the long term, the strongest cumulative decline appears in Financials and Construction in terms of magnitude, while **News & Publishing** is the only industry with a statistically significant cumulative loss.
  These results suggest that some industries experience persistent immediate shocks, whereas others adjust more gradually over several days.

Analysis 02

Return

Volatility

Volume

Volatility Calculations

Short Term [-1h, +1h]

Pre-Event Window (-60 to -1 min):

$$RV_{pre} = \sum_{m=-60}^{-1} r_m^2 \ * \ 1/60$$

Post-Event Window (0 to +60 min):

$$RV_{post} = \sum_{m=0}^{+60} r_m^2 \ * \ 1/61$$

Delta RV:

$$\Delta RV = RV_{post} - RV_{pre}$$

Industry AAV:

$$AAV_t^{(ind)} = \frac{1}{N_{ind}} \sum_{e=1}^{N_{ind}} \left(r_{e,t}^2 - \frac{RV_{pre,e}}{60} \right)$$

Long Term [-5d, +5d]

Pre-Event Window (-5 to -1 days):

$$Pre_{RV} = \frac{1}{5} \sum_{d=-5}^{-1} r_d^2$$

Post-Event Window (+1 to +5 days):

$$Post_{RV} = \frac{1}{5} \sum_{d=+1}^{+5} r_d^2$$

Delta RV:

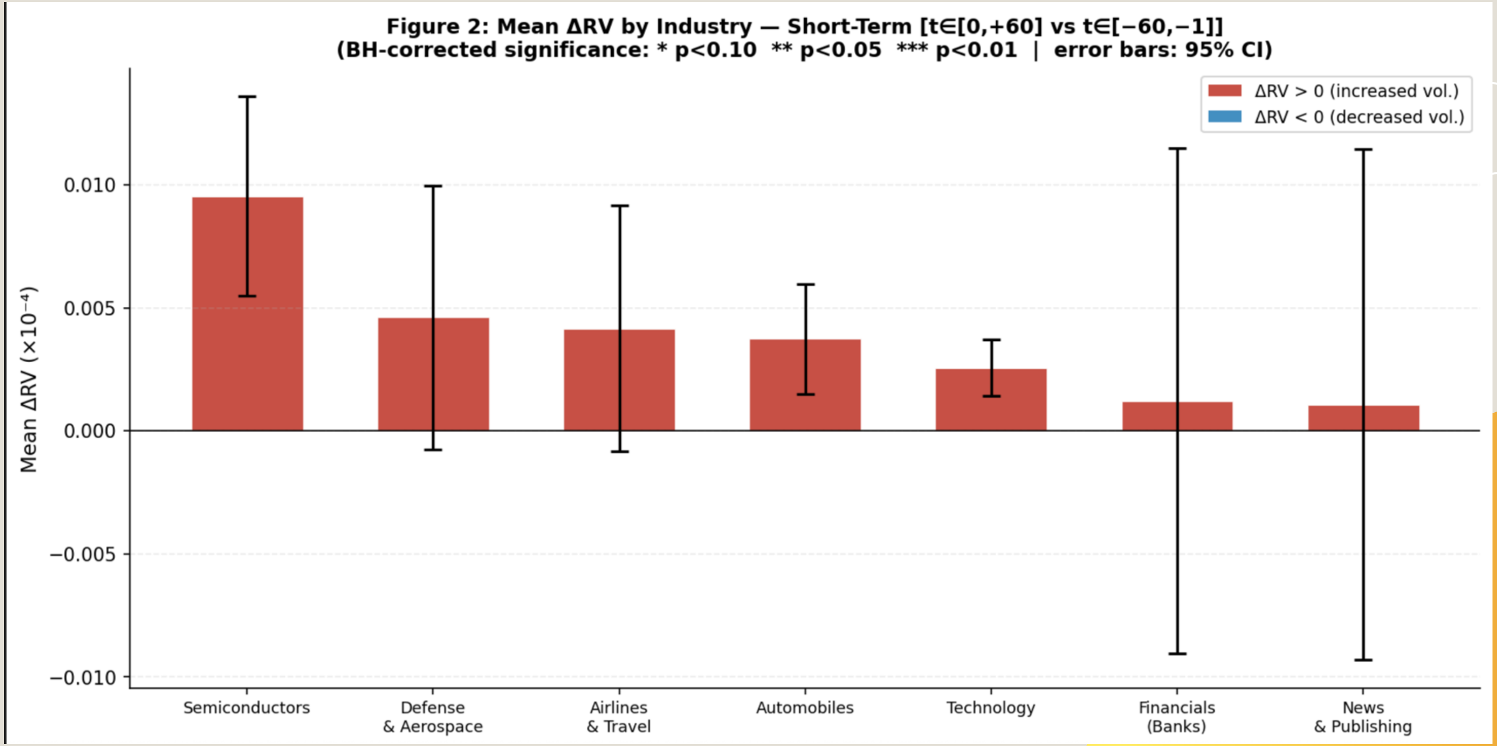
$$\Delta RV = Post_{RV} - Pre_{RV}$$

Industry AAV:

$$AAV_d^{(ind)} = \frac{1}{N_{ind}} \sum_{e=1}^{N_{ind}} (r_{e,d}^2 - RV_{pre,e})$$

Short-Term ARV: Significant First-Hour Volatility Jump with Cross-Industry Divergence

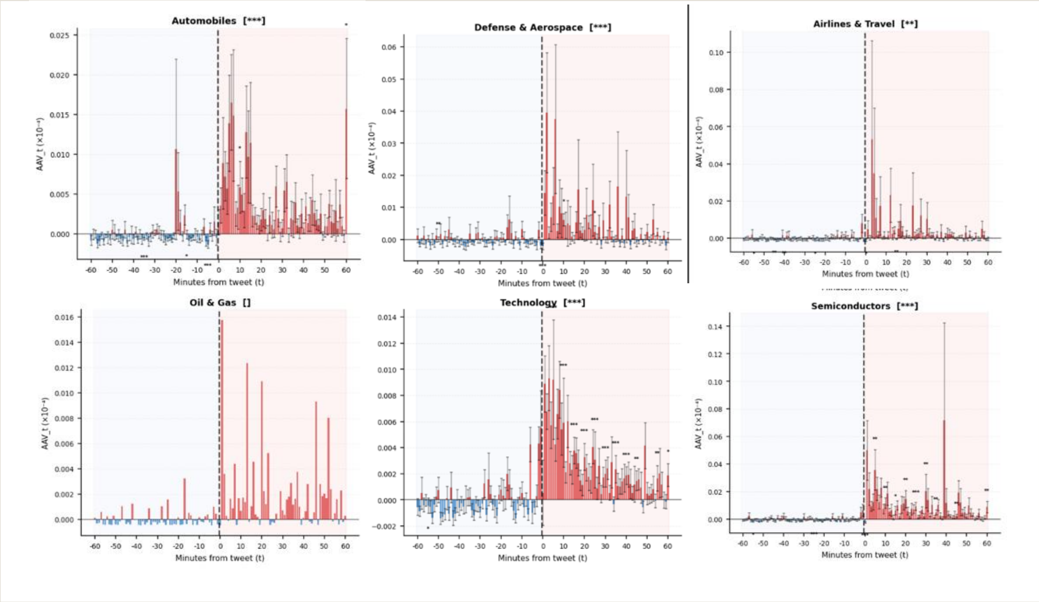
In the **first hour** after the tweet, **most industries** exhibit a significant **increase** in volatility, with **Semiconductors** showing the strongest jump. **Financials and News & Publishing** display positive but statistically insignificant responses, highlighting clear cross-industry heterogeneity in short-term volatility effects.



Short-term AAV: Divergent Intraday Volatility Response to Trump's Tweets

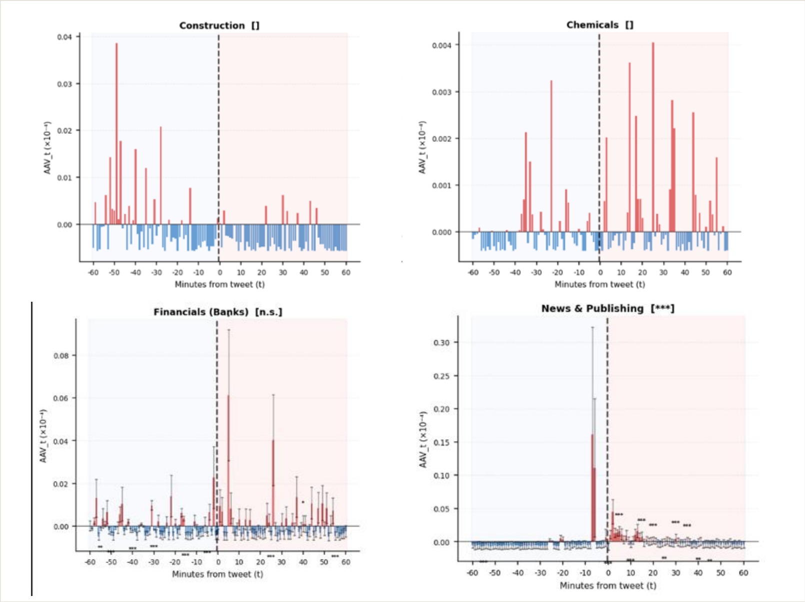
1. Strong Positive Responders

The strongest intraday reactions concentrate in **high-tech, information-sensitive sectors and in cyclical industries**, exhibit clear and statistically significant jumps, heavily exposed to trade and policy risk.



2.Weaker or Unclear Responses

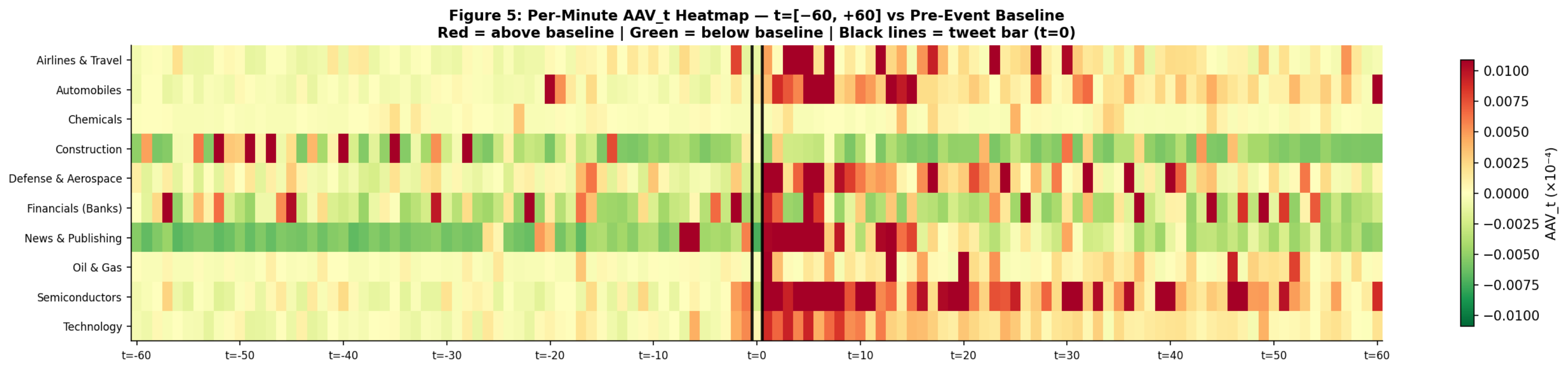
Financials, Construction, Chemicals, News & Publishing show limited, inconsistent, or statistically insignificant intraday reactions, suggesting that the shock is not uniformly transmitted across industries.



Short-term AAV: Intraday Volatility Clustering Around Tweet Announcements

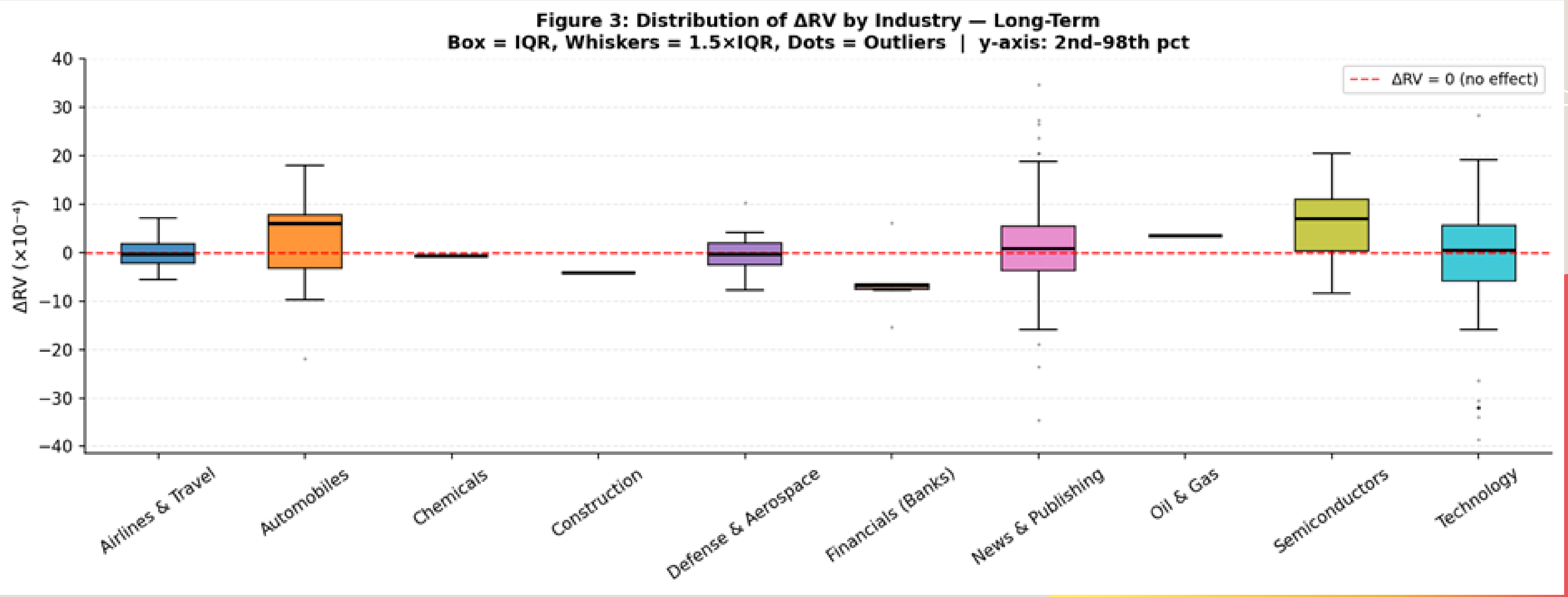
Timing Pattern:

Volatility spikes **sharply at t=1** and remains elevated for approximately **10–20 minutes,** indicating a rapid market reaction followed by gradual normalization.



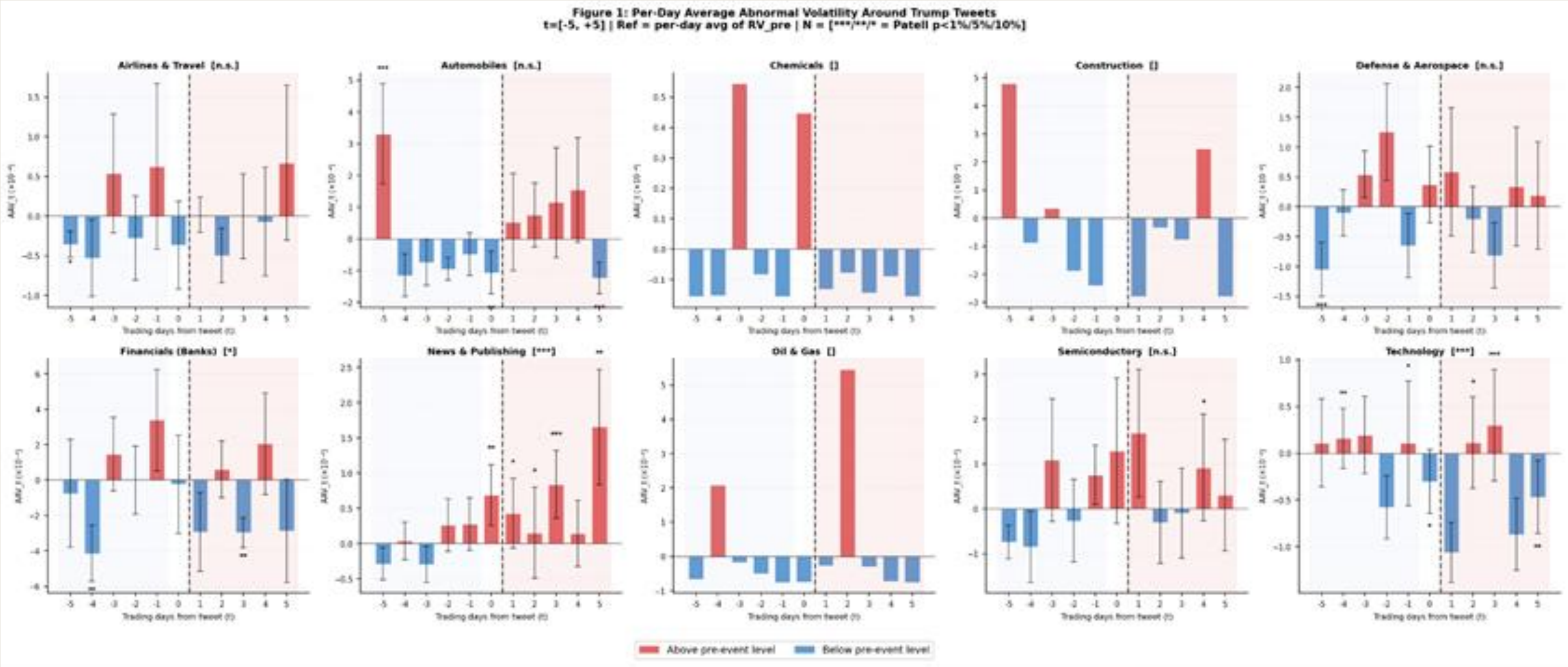
Long-term : Overall ΔRV Distribution differs between industries

- Several industries show a **positive shift** in median ΔRV (e.g., Semiconductors, Automobiles, News & Publishing).
- Financials exhibit a **negative** median shift.
- However, distributions overlap zero substantially, suggesting limited systematic persistence.



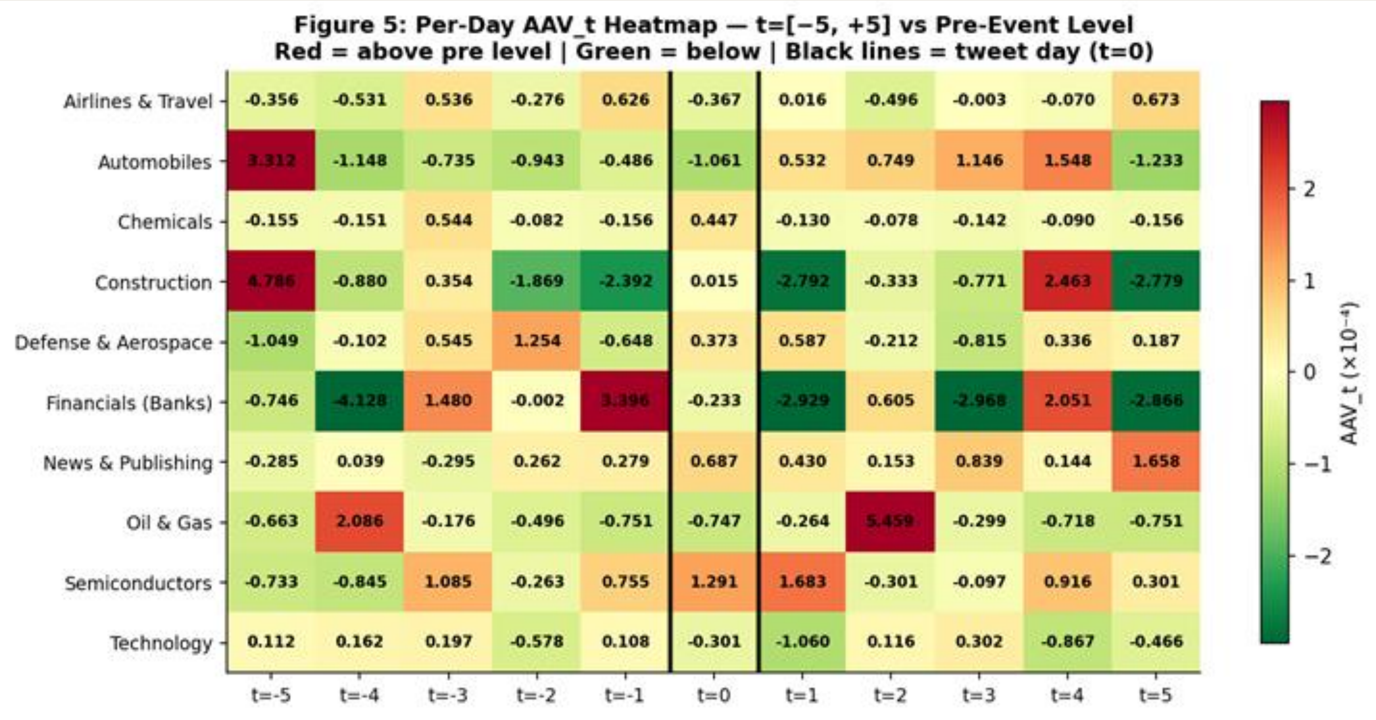
Long-Term AAV: Volatility Responses Lack Systematic Persistence

- Long-term volatility responses are heterogeneous and often **preceded by significant pre-event movements.**
- Only **News & Publishing** showing a modest positive effect and **Financials** exhibiting a weak negative adjustment. **Automobiles** showing temporary post-tweet increases rather than sustained structural shifts.



Long-term AAV: Heterogeneous and Transitory Daily Volatility Adjustments

- Daily abnormal volatility patterns **appear dispersed across both pre- and post-event windows.**
- Several industries exhibit notable movements prior to the tweet, while post-event adjustments are episodic rather than persistent, indicating limited long-term structural impact.



Findings 02

Volatility

Short-term

- Trump's firm-specific tweets trigger strongest intraday reactions **concentrate in high-tech, information-sensitive sectors and in cyclical industries.**

Long-term

- Volatility Responses **Lack Systematic Persistence**
- Only **News & Publishing** showing a modest **positive** effect and **Financials** exhibiting a **weak negative** adjustment.

4.Discussion & Appendix

Hypotheses Conclusion

	Hypotheses	Findings
general	Trump's tweets generate significant abnormal influence on stock returns in related industries.	Supported. Significant abnormal returns appear in **Financials and Semiconductors** in the short term, while **News & Publishing shows the only significant long-term loss**.
SUB-H1	Short-term effects are stronger than long-term effects.Over time, the impact largely reverts to normal levels.	Supported. Return and volatility effects are strongest immediately after the tweet and largely fade over time, with persistence mainly in **News & Publishing**.
SUB-H2	The impact is mainly concentrated in the short term, and significantly differs across industries.	Strongly supported. **Semiconductors and Technology react positively**, while **Chemicals and News react negatively**.

Limitations

1

Multi-factors

Causal inference is limited by anticipation effects, tweet clustering, and uncontrolled sentiment factors.

2

Small sample sizes

Results for Chemicals, Construction, and Oil & Gas (N=1) are statistically unreliable, and Financials (N=6) cannot reach significance despite its large effect size.

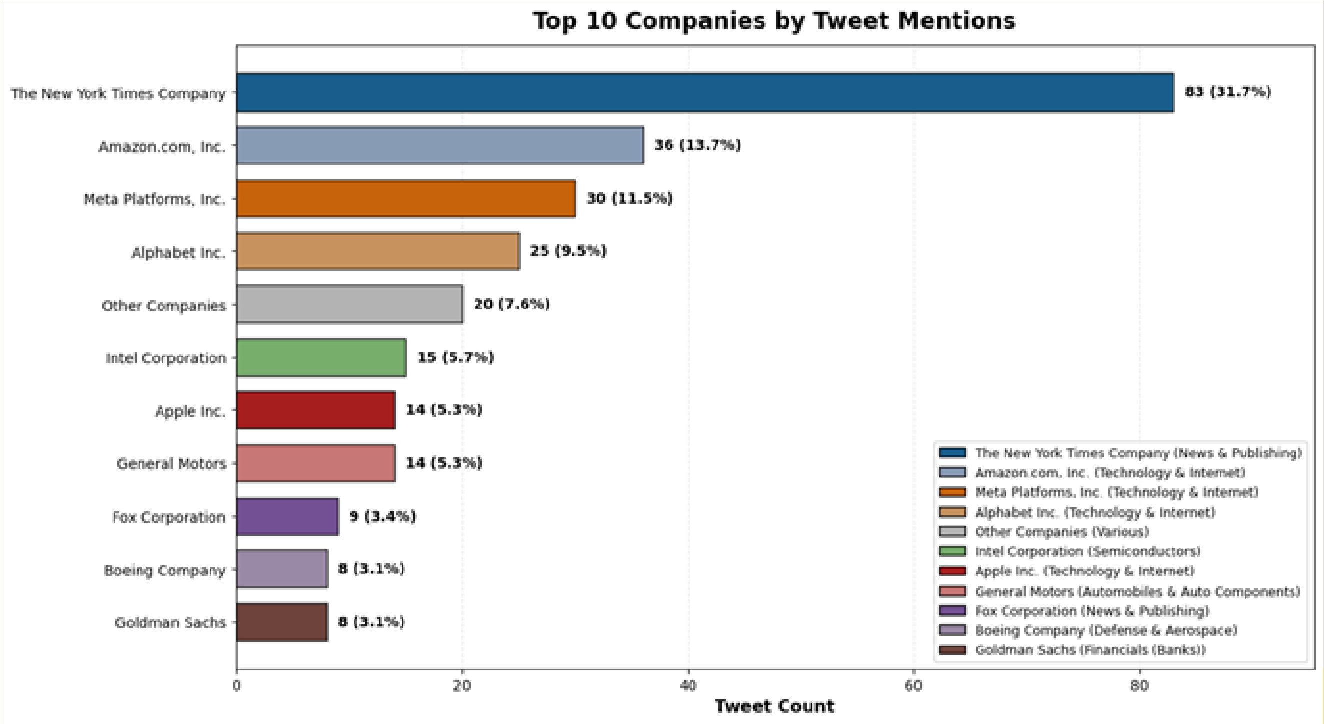
3

Tweet Matching Precision

Matching tweets using a predefined dictionary may cause misclassification (false positives/negatives), which can bias abnormal return estimates and affect causal inference.

Appendix

Company and Industry Distribution



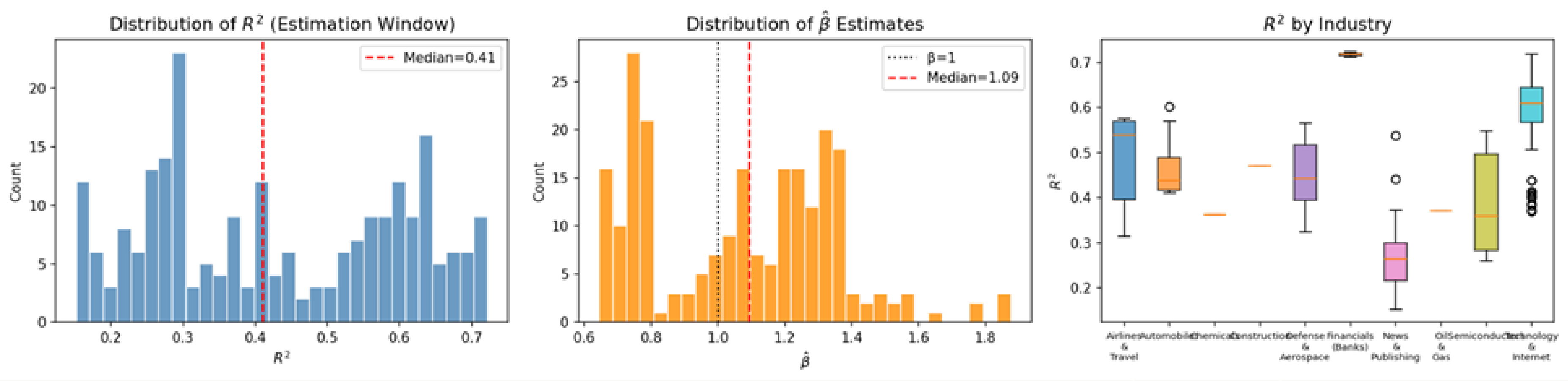
Strong firm-level concentration

Mentions are highly concentrated. The New York Times Company dominates, followed by Amazon, Meta, and Alphabet.

Higher political exposure for media and tech firms

Technology & Internet and News & Publishing receive the highest total mentions, suggesting greater political exposure and attention intensity in these sectors.

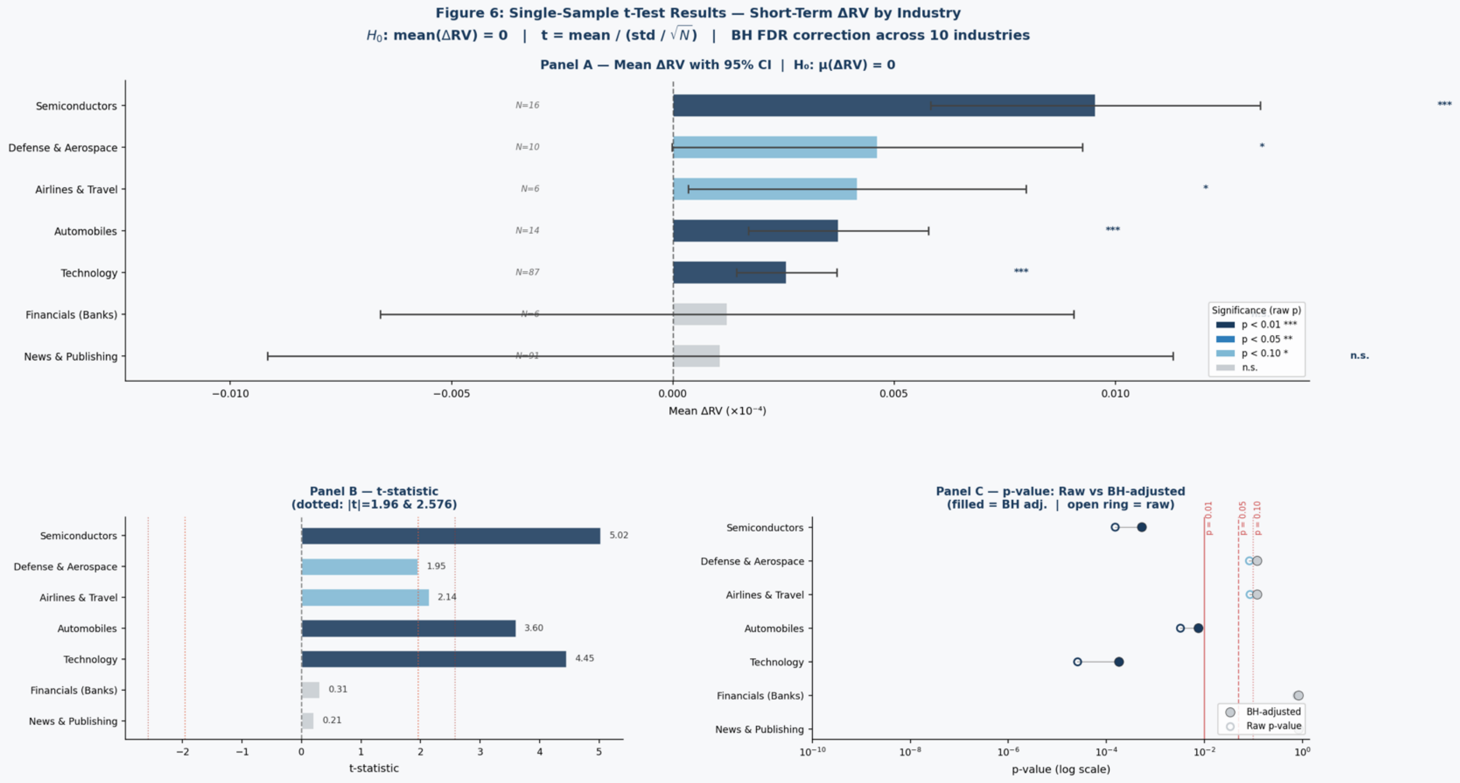
Return：CAPM Market Model: Estimation Quality



Median R²=0.41 and β=1.09 confirm reliable model estimation; note that Chemicals, Construction, and Oil & Gas each have only one @-mentioned company, limiting the reliability of their estimates.

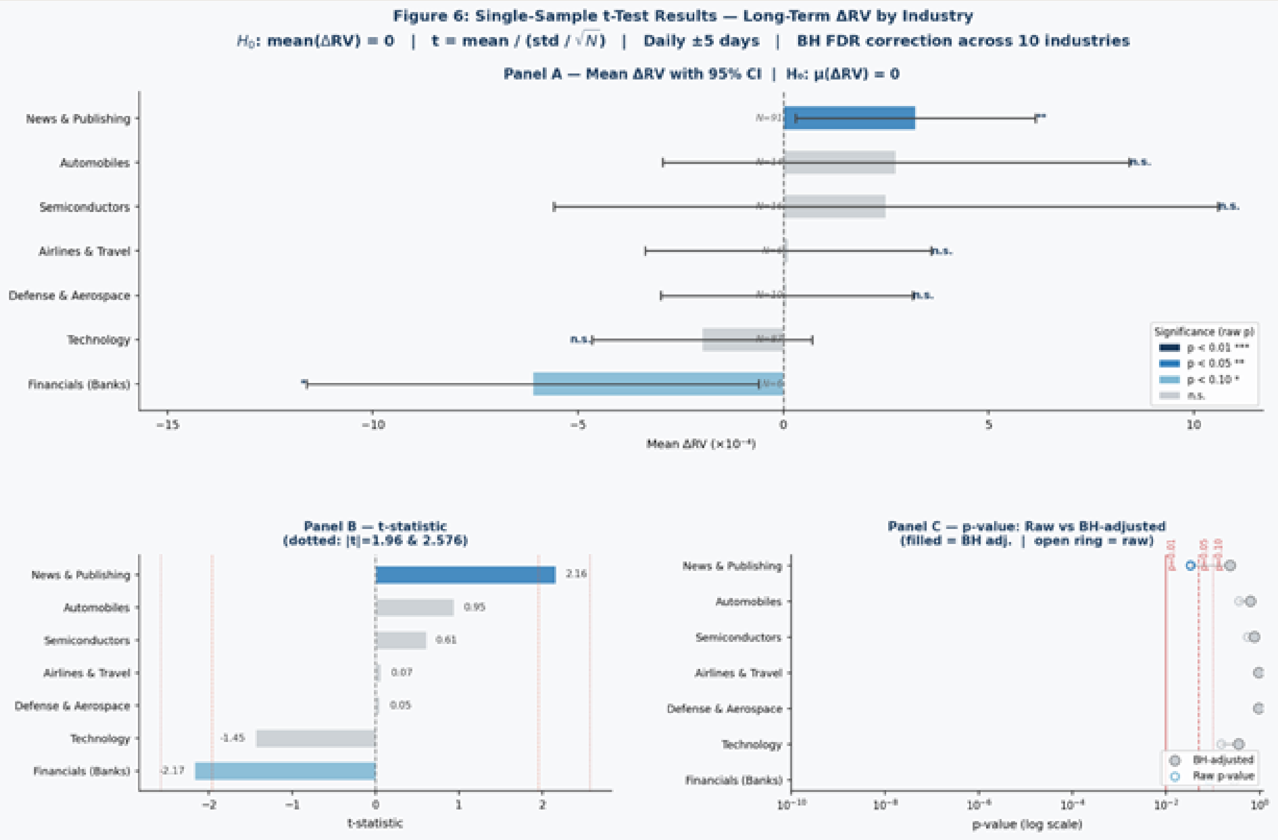
Short-term: Volatility Test

Short-term ΔRV is significantly positive in Semiconductors, Technology, Automobiles, Defense, and Airlines, and these results remain robust after BH multiple-testing correction.



Long-term: Volatility Test

Long-term volatility effects are largely insignificant across industries, with only News & Publishing showing a modest positive effect and Financials exhibiting a weak negative adjustment.



THANKS FOR YOUR ATTENTION!!!